Counting a Set and Telling How Many
Answer Key

Kindergarten–Grade 1 Counting and Cardinality

Quick Answers

1. 7	4. 9	7. 13	10. —
2. 9	5. 8	8. 16	
3. 12	6. —	9. 9	

Common wrong answers

1. *If they got 8: touched one star twice while counting, so the count ran one past the set*
2. *If they got 8: skipped a counter that was set apart from the others, so the count came up one short*
4. *If they got 5: counted only the larger group and forgot the other one*
9. *If they got 7: counted only the first two baskets and stopped*

1. Stars in the ten-frame.

Step 1. The top row of a ten-frame holds five, and this top row is full, so the count reaches 5 at the end of that row.
Step 2. Count on into the bottom row, touching one star per number: "six, seven". That is $5 + 2$.
Step 3. The last number said is the answer to *how many*, so the set has $\boxed{7}$ stars.
Listen for: a child who recounts the top row from one when moving to the second row. *Common wrong answer:* 8 — one star was touched twice.

2. Scattered counters.

Step 1. Scattered objects are the classic place to lose track, so give the set an order: count the upper band first, left to right — 5 counters.
Step 2. Continue into the lower band without restarting: "six, seven, eight, nine". That band holds 4, and $5 + 4$ is the whole set.
Step 3. Every counter was touched exactly once, so the set has $\boxed{9}$ counters.
Common wrong answer: 8 — one counter was skipped.

3. Full ten-frame and two loose dots.

Step 1. A full ten-frame is 10. Because it is full, it does not have to be recounted — that is the whole point of the frame.
Step 2. Start at ten and count on for the two loose dots: "eleven, twelve". That is $10 + 2$.
Step 3. The last number said is $\boxed{12}$ dots.
Teaching note: counting on from a known amount is the first step toward place value — twelve is "ten and two more".

4. Ducks in two places.

Step 1. The ducks are one set shown in two places, so count the pond first: 4 ducks.
Step 2. Keep the count going onto the grass: "five, six, seven, eight, nine". That is $4 + 5$.
Step 3. In all there are $\boxed{9}$ ducks.
Common wrong answer: 5 — only the grass group was counted.

5. Buttons in uneven rows.

Step 1. Rows of different lengths tempt a child to assume they match, so count each row and keep a running total instead of guessing.
Step 2. Row by row: 3, then 3 more reaches six, then 2 more. That is $3 + 3 + 2$.
Step 3. The last number said is $\boxed{8}$ buttons.

6. Rosa's shells and Milo's shells.

Step 1. Comparing needs two finished counts, so count Rosa first: 5 shells in her first group and 3 in her second, which is $5 + 3 = 8$.
Step 2. Now Milo: $4 + 2 = 6$ shells.
Step 3. Eight is more than six, so the true sentence uses the *greater than* sign: Rosa's shells $\boxed{>}$ Milo's shells.
Listen for: "Milo has more because his shells are spread out." The count decides, not how much space the set takes up.

7. Red and blue blocks.

Step 1. Both colours are blocks in the same box, so one count covers both groups.
Step 2. Count the red blocks: 7. Count on through the blue ones: "eight, nine, ten, eleven, twelve, thirteen" — that is $7 + 6$.
Step 3. The box holds $\boxed{13}$ blocks.

8. Full muffin tray.

Step 1. The tray is full, so the rows are equal — count one row and then continue through the second row.
Step 2. The first row holds 8. Counting on through the second row gives "nine, ten, . . . , sixteen", which is $8 + 8$.
Step 3. There are $\boxed{16}$ muffins.

9. Three baskets of apples.

Step 1. Three groups still make one set, so the count must run straight through all three baskets without restarting.
Step 2. Basket A holds 4; carry on into basket B — "five, six, seven" — which adds 3; carry on into basket C — "eight, nine" — which adds 2. That is $4 + 3 + 2$.
Step 3. There are $\boxed{9}$ apples in all.
Common wrong answer: 7 — the third basket was never counted.

10. Two jars of marbles.

Step 1. Each ten-frame is full, so each jar starts at 10 and only the extra marbles need counting.

Step 2. Jar A: count on from ten for four extras — "eleven, twelve, thirteen, fourteen", so $10 + 4 = 14$. Jar B: count on for six extras, so $10 + 6 = 16$.

Step 3. Fourteen is less than sixteen, so the true sentence uses the *less than* sign: Jar A $\boxed{<}$ Jar B.

Teaching note: because both jars share the same full ten-frame, only the loose marbles can decide the comparison — a useful preview of comparing teen numbers by their ones digit.